

Module 1: Basic programming in R

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Methods and computing camp

This summer we will together learn and review materials of statistical computing and methods.

What do we do during lectures?

Materials will be available at course website. Lecture notes are created by Rmarkdown.

- **If you have questions, feel free to interrupt or send a message in the chat.**

We will cover 5 modules of statistical methods and computing.

- Each module takes ~1.75 hours.

What contents we will cover?

Module	Topics	References
1	Tidyverse basics; data wrangling & graphing; Using LLMs to generate R code	-
2	Statistical inference (I)	AoS Chp 1-5; AoS Chp 6; C&B Chp 6.3, 7
3	Statistical inference (II)	AoS Chp 8; C&B Chp 8
4	Linear Regression & Generalized Linear Models	AoS Chp 13; C&B Chp 11-12
5	Simulation and parallel computing	C&B Chap 10; AoS Chp 5, 24;

What contents we will not cover?

- Nonparametric inference (STA3000)
- Bayesian inference (STA3000, STA2201)
- Computing of Bayesian inference (STA2201)
- Shiny and blogdown in R

Exercises!

- Available for each module.
- Exercises can be difficult.
- Group discussion is recommended.

Prerequisites:

- Students are expected to have some basic experience with R, Rmarkdown and LaTeX.
- See Module 0 for some background

Code style

- [Tidyverse style guide](#)
- `styler` software embedded in Rstudio - allows you to interactively restyle selected text, files, or entire projects.
- `lintr` software embedded in Rstudio - performs automated checks to confirm that you conform to the style guide.

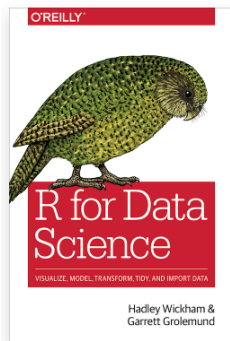
Outline

We will review R, Rstudio, and Syntax of R together.

- Tidy data, processing (tidyverse)
- Graphing (ggplot2)

Resources

- “R for Data Science: Import, Tidy, Transform, Visualize, and Model Data” by Hadley Wickham.
- Brendan R. E. Ansell’s “Introduction to R - tidyverse” [\[link\]](#)



Data import

```
df <- read.table("mtcars.txt", header = TRUE)
head(df) # Show the first 6 rows.
```

```
##   Cntry lper100k weight length
## 1    US      19.8   2178   5.92
## 2  Japan      9.9   1026   4.32
## 3    US     10.8   1188   4.27
## 4    US     12.5   1444   5.11
## 5    US     12.5   1485   5.03
## 6    US     12.5   1485   5.03
```

Other options

CSV files.

- `base::read.csv()` in the base r.
- `readr::read.csv()` in “readr” package (much faster).
- `data.table::fread()` in “data.table” package (much more faster).

Rdata.

- `load()` in the base r.

Tidy data

The goal is to clean the dataset so it is much easier to use.

Specifically,

- Each variable must have its own column.
- Each observation must have its own row.
- Each value must have its own cell.

We will focus on the functions from `tidyverse` package.

```
library(tidyverse)
```

Tidy data 1: pivoting

For a dataset having column names are not names of variables, but values of a variable, e.g.

```
# table4a documents the number  
# of documented TB cases in  
# Afghanistan, Brazil and China  
table4a
```

```
## # A tibble: 3 x 3  
##   country   `1999` `2000`  
##   <chr>     <dbl> <dbl>  
## 1 Afghanistan    745   2666  
## 2 Brazil        37737  80488  
## 3 China         212258 213766
```

- Need to change 1999, 2000 to a column? named as "year".
- Need to change the values of 1999, 2000 as "cases".

We can use `pivot_longer()` from the "tidyverse" package.

Pivot longer

```
table4a %>%  
  pivot_longer(c(`1999`, `2000`),  
    names_to = "year", values_to = "cases")
```

```
## # A tibble: 6 x 3  
##   country    year  cases  
##   <chr>      <chr> <dbl>  
## 1 Afghanistan 1999    745  
## 2 Afghanistan 2000   2666  
## 3 Brazil      1999  37737  
## 4 Brazil      2000  80488  
## 5 China       1999 212258  
## 6 China       2000 213766
```

Another example

```
table2 %>% head(5)
```

```
## # A tibble: 5 x 4
##   country      year type      count
##   <chr>      <dbl> <chr>    <dbl>
## 1 Afghanistan 1999 cases      745
## 2 Afghanistan 1999 population 19987071
## 3 Afghanistan 2000 cases      2666
## 4 Afghanistan 2000 population 20595360
## 5 Brazil      1999 cases      37737
```

- case and population are two variables and should be converted into columns.

We can use `pivot_wider()`.

Pivot wider

```
table2 %>%  
  pivot_wider(names_from = type, values_from = count)
```

```
## # A tibble: 6 x 4  
##   country      year  cases population  
##   <chr>      <dbl> <dbl>      <dbl>  
## 1 Afghanistan 1999     745   19987071  
## 2 Afghanistan 2000    2666  20595360  
## 3 Brazil      1999   37737  172006362  
## 4 Brazil      2000   80488  174504898  
## 5 China       1999  212258  1272915272  
## 6 China       2000  213766  1280428583
```

Transform data

Use the “pipes” from the “tidyverse” package, a powerful tool for clearly expressing a sequence of multiple operations, with the combination of the following functions:

- `select()`
- `filter()`
- `arrange()`
- `mutate()`
- `summarise()`
- `group_by()`

Dataset - Diamonds

A dataset containing the prices and other attributes of almost 54,000 diamonds.

```
head(diamonds)
```

```
## # A tibble: 6 x 10
##   carat cut      color clarity depth table price      x      y      z
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>
## 1  0.23 Ideal      E     SI2      61.5   55   326   3.95   3.98   2.43
## 2  0.21 Premium    E     SI1      59.8   61   326   3.89   3.84   2.31
## 3  0.23 Good       E     VS1      56.9   65   327   4.05   4.07   2.31
## 4  0.29 Premium    I     VS2      62.4   58   334   4.2    4.23   2.63
## 5  0.31 Good       J     SI2      63.3   58   335   4.34   4.35   2.75
## 6  0.24 Very Good J     VVS2      62.8   57   336   3.94   3.96   2.48
```

Select

Use `select()` to get a column, e.g. “color”

```
diamonds %>%  
  select(color) %>%  
  head()
```

```
## # A tibble: 6 x 1  
##   color  
##   <ord>  
## 1 E  
## 2 E  
## 3 E  
## 4 I  
## 5 J  
## 6 J
```

```
# Equivalent to...  
head(diamonds$color)
```

```
## [1] E E E I J J  
## Levels: D < E < F < G < H < I < J
```

Select

Use `select()` to remove a column, e.g. “color”

```
diamonds %>%  
  select(-color)
```

```
## # A tibble: 53,940 x 9  
##   carat cut      clarity depth table price     x     y     z  
##   <dbl> <ord>    <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.23 Ideal    SI2      61.5   55   326   3.95   3.98   2.43  
## 2  0.21 Premium SI1      59.8   61   326   3.89   3.84   2.31  
## 3  0.23 Good    VS1      56.9   65   327   4.05   4.07   2.31  
## 4  0.29 Premium VS2      62.4   58   334   4.2    4.23   2.63  
## 5  0.31 Good    SI2      63.3   58   335   4.34   4.35   2.75  
## 6  0.24 Very Good VVS2    62.8   57   336   3.94   3.96   2.48  
## 7  0.24 Very Good VVS1    62.3   57   336   3.95   3.98   2.47  
## 8  0.26 Very Good SI1      61.9   55   337   4.07   4.11   2.53  
## 9  0.22 Fair    VS2      65.1   61   337   3.87   3.78   2.49  
## 10 0.23 Very Good VS1      59.4   61   338   4     4.05   2.39  
## # i 53,930 more rows
```

```
# Need to assign the change to the original dataset, otherwise, the deletion won't affect the dataset.  
diamonds <- diamonds %>%  
  select(-color)
```

Filter

Use `filter()` to filter by some condition, e.g. filter all price > 335

```
diamonds %>%  
  filter(price > 335)
```

```
## # A tibble: 53,935 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.24 Very Good J      VVS2    62.8    57    336    3.94    3.96    2.48  
## 2  0.24 Very Good I      VVS1    62.3    57    336    3.95    3.98    2.47  
## 3  0.26 Very Good H      SI1     61.9    55    337    4.07    4.11    2.53  
## 4  0.22 Fair      E      VS2     65.1    61    337    3.87    3.78    2.49  
## 5  0.23 Very Good H      VS1     59.4    61    338    4      4.05    2.39  
## 6  0.3   Good      J      SI1     64      55    339    4.25    4.28    2.73  
## 7  0.23 Ideal     J      VS1     62.8    56    340    3.93    3.9      2.46  
## 8  0.22 Premium   F      SI1     60.4    61    342    3.88    3.84    2.33  
## 9  0.31 Ideal     J      SI2     62.2    54    344    4.35    4.37    2.71  
## 10 0.2   Premium   E      SI2     60.2    62    345    3.79    3.75    2.27  
## # i 53,925 more rows
```

Filters with multiple conditions

```
diamonds %>%  
  filter(price > 335 & depth < 64)
```

```
## # A tibble: 51,849 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.24 Very Good J      VVS2    62.8    57    336  3.94  3.96  2.48  
## 2  0.24 Very Good I      VVS1    62.3    57    336  3.95  3.98  2.47  
## 3  0.26 Very Good H      SI1     61.9    55    337  4.07  4.11  2.53  
## 4  0.23 Very Good H      VS1     59.4    61    338   4     4.05  2.39  
## 5  0.23 Ideal    J      VS1     62.8    56    340  3.93  3.9   2.46  
## 6  0.22 Premium  F      SI1     60.4    61    342  3.88  3.84  2.33  
## 7  0.31 Ideal    J      SI2     62.2    54    344  4.35  4.37  2.71  
## 8  0.2  Premium  E      SI2     60.2    62    345  3.79  3.75  2.27  
## 9  0.32 Premium  E      I1      60.9    58    345  4.38  4.42  2.68  
## 10 0.3  Ideal    I      SI2     62     54    348  4.31  4.34  2.68  
## # i 51,839 more rows
```

Filters with multiple conditions

```
diamonds %>%  
  filter(cut == "Very Good" | cut == "Fair")
```

```
## # A tibble: 13,692 x 10
```

##	carat	cut	color	clarity	depth	table	price	x	y	z
##	<dbl>	<ord>	<ord>	<ord>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
## 1	0.24	Very Good	J	VVS2	62.8	57	336	3.94	3.96	2.48
## 2	0.24	Very Good	I	VVS1	62.3	57	336	3.95	3.98	2.47
## 3	0.26	Very Good	H	SI1	61.9	55	337	4.07	4.11	2.53
## 4	0.22	Fair	E	VS2	65.1	61	337	3.87	3.78	2.49
## 5	0.23	Very Good	H	VS1	59.4	61	338	4	4.05	2.39
## 6	0.3	Very Good	J	SI1	62.7	59	351	4.21	4.27	2.66
## 7	0.23	Very Good	E	VS2	63.8	55	352	3.85	3.92	2.48
## 8	0.23	Very Good	H	VS1	61	57	353	3.94	3.96	2.41
## 9	0.31	Very Good	J	SI1	59.4	62	353	4.39	4.43	2.62
## 10	0.31	Very Good	J	SI1	58.1	62	353	4.44	4.47	2.59

```
## # i 13,682 more rows
```

Filter after select

This is an example of “a sequence of operations”.

```
diamonds %>%  
  select(price) %>%  
  filter(price > 335)
```

```
## # A tibble: 53,935 x 1  
##   price  
##   <int>  
## 1    336  
## 2    336  
## 3    337  
## 4    337  
## 5    338  
## 6    339  
## 7    340  
## 8    342  
## 9    344  
## 10   345  
## # i 53,925 more rows
```

Arrange

Use `arrange()` to order data.

```
diamonds %>%  
  arrange(price)
```

```
## # A tibble: 53,940 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.23 Ideal    E     SI2     61.5   55   326   3.95   3.98   2.43  
## 2  0.21 Premium E     SI1     59.8   61   326   3.89   3.84   2.31  
## 3  0.23 Good     E     VS1     56.9   65   327   4.05   4.07   2.31  
## 4  0.29 Premium I     VS2     62.4   58   334   4.2    4.23   2.63  
## 5  0.31 Good     J     SI2     63.3   58   335   4.34   4.35   2.75  
## 6  0.24 Very Good J     VVS2    62.8   57   336   3.94   3.96   2.48  
## 7  0.24 Very Good I     VVS1    62.3   57   336   3.95   3.98   2.47  
## 8  0.26 Very Good H     SI1     61.9   55   337   4.07   4.11   2.53  
## 9  0.22 Fair     E     VS2     65.1   61   337   3.87   3.78   2.49  
## 10 0.23 Very Good H     VS1     59.4   61   338   4      4.05   2.39  
## # i 53,930 more rows
```

Arrange descending order

e.g. from the cheapest!

```
diamonds %>%  
  arrange(-price)
```

```
## # A tibble: 53,940 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  2.29 Premium  I     VS2     60.8   60 18823  8.5   8.47  5.16  
## 2  2      Very Good G     SI1     63.5   56 18818  7.9   7.97  5.04  
## 3  1.51 Ideal    G     IF      61.7   55 18806  7.37  7.41  4.56  
## 4  2.07 Ideal    G     SI2     62.5   55 18804  8.2   8.13  5.11  
## 5  2      Very Good H     SI1     62.8   57 18803  7.95  8     5.01  
## 6  2.29 Premium  I     SI1     61.8   59 18797  8.52  8.45  5.24  
## 7  2.04 Premium  H     SI1     58.1   60 18795  8.37  8.28  4.84  
## 8  2      Premium  I     VS1     60.8   59 18795  8.13  8.02  4.91  
## 9  1.71 Premium  F     VS2     62.3   59 18791  7.57  7.53  4.7  
## 10 2.15 Ideal    G     SI2     62.6   54 18791  8.29  8.35  5.21  
## # i 53,930 more rows
```

Arrange by multiple conditions

```
diamonds %>%  
  arrange(price, cut)
```

```
## # A tibble: 53,940 x 10
```

##	carat	cut	color	clarity	depth	table	price	x	y	z
##	<dbl>	<ord>	<ord>	<ord>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
## 1	0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
## 2	0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
## 3	0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
## 4	0.29	Premium	I	VS2	62.4	58	334	4.2	4.23	2.63
## 5	0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
## 6	0.24	Very Good	J	VVS2	62.8	57	336	3.94	3.96	2.48
## 7	0.24	Very Good	I	VVS1	62.3	57	336	3.95	3.98	2.47
## 8	0.22	Fair	E	VS2	65.1	61	337	3.87	3.78	2.49
## 9	0.26	Very Good	H	SI1	61.9	55	337	4.07	4.11	2.53
## 10	0.23	Very Good	H	VS1	59.4	61	338	4	4.05	2.39

```
## # i 53,930 more rows
```

Filter, select, arrange

```
diamonds %>%  
  filter(table < 340) %>%  
  select(carat, cut, price) %>%  
  arrange(price, cut)
```

```
## # A tibble: 53,940 x 3  
##   carat cut      price  
##   <dbl> <ord>    <int>  
## 1  0.21 Premium     326  
## 2  0.23 Ideal      326  
## 3  0.23 Good       327  
## 4  0.29 Premium     334  
## 5  0.31 Good       335  
## 6  0.24 Very Good   336  
## 7  0.24 Very Good   336  
## 8  0.22 Fair       337  
## 9  0.26 Very Good   337  
## 10 0.23 Very Good   338  
## # i 53,930 more rows
```

Mutate

Create new variables using `mutate()`.

- Create a boolean variable, 0 = not affordable, 1 = affordable.

```
diamonds %>%  
  mutate(affordable = price < 400)
```

```
## # A tibble: 53,940 x 11  
##   carat cut      color clarity depth table price      x      y      z affordable  
##   <dbl> <ord>      <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl> <lgl>  
## 1  0.23 Ideal      E      SI2     61.5    55   326   3.95   3.98   2.43 TRUE  
## 2  0.21 Premium    E      SI1     59.8    61   326   3.89   3.84   2.31 TRUE  
## 3  0.23 Good       E      VS1     56.9    65   327   4.05   4.07   2.31 TRUE  
## 4  0.29 Premium    I      VS2     62.4    58   334   4.2    4.23   2.63 TRUE  
## 5  0.31 Good       J      SI2     63.3    58   335   4.34   4.35   2.75 TRUE  
## 6  0.24 Very Good J      VVS2     62.8    57   336   3.94   3.96   2.48 TRUE  
## 7  0.24 Very Good I      VVS1     62.3    57   336   3.95   3.98   2.47 TRUE  
## 8  0.26 Very Good H      SI1     61.9    55   337   4.07   4.11   2.53 TRUE  
## 9  0.22 Fair       E      VS2     65.1    61   337   3.87   3.78   2.49 TRUE  
## 10 0.23 Very Good H      VS1     59.4    61   338   4      4.05   2.39 TRUE  
## # i 53,930 more rows
```

Mutate (cont'd)

- Create a variable containing string with `case_when()`:

```
diamonds %>%  
  mutate(affordable = case_when(price<400 ~ "affordable",  
                                TRUE ~ "not affordable"))
```

```
## # A tibble: 53,940 x 11  
##   carat cut      color clarity depth table price      x      y      z affordable  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl> <chr>  
## 1  0.23 Ideal      E      SI2     61.5    55   326   3.95   3.98   2.43 affordable  
## 2  0.21 Premium    E      SI1     59.8    61   326   3.89   3.84   2.31 affordable  
## 3  0.23 Good       E      VS1     56.9    65   327   4.05   4.07   2.31 affordable  
## 4  0.29 Premium    I      VS2     62.4    58   334   4.2    4.23   2.63 affordable  
## 5  0.31 Good       J      SI2     63.3    58   335   4.34   4.35   2.75 affordable  
## 6  0.24 Very Good J      VVS2     62.8    57   336   3.94   3.96   2.48 affordable  
## 7  0.24 Very Good I      VVS1     62.3    57   336   3.95   3.98   2.47 affordable  
## 8  0.26 Very Good H      SI1     61.9    55   337   4.07   4.11   2.53 affordable  
## 9  0.22 Fair       E      VS2     65.1    61   337   3.87   3.78   2.49 affordable  
## 10 0.23 Very Good H      VS1     59.4    61   338   4      4.05   2.39 affordable  
## # i 53,930 more rows
```

Group by and Summarise

Use `group_by` and `summarise` to group variables:

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n())
```

```
## # A tibble: 5 x 2  
##   cut      n  
##   <ord>   <int>  
## 1 Fair    1610  
## 2 Good    4906  
## 3 Very Good 12082  
## 4 Premium 13791  
## 5 Ideal   21551
```

More examples

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price))
```

```
## # A tibble: 5 x 3  
##   cut          n price_avg  
##   <ord>      <int>      <dbl>  
## 1 Fair        1610      4359.  
## 2 Good        4906      3929.  
## 3 Very Good  12082      3982.  
## 4 Premium    13791      4584.  
## 5 Ideal      21551      3458.
```

Proportions

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price)) %>%  
  ungroup() %>%  
  mutate(prop = n/sum(n))
```

```
## # A tibble: 5 x 4  
##   cut          n price_avg  prop  
##   <ord>    <int>    <dbl> <dbl>  
## 1 Fair      1610     4359. 0.0298  
## 2 Good      4906     3929. 0.0910  
## 3 Very Good 12082     3982. 0.224  
## 4 Premium   13791     4584. 0.256  
## 5 Ideal     21551     3458. 0.400
```

With percentage

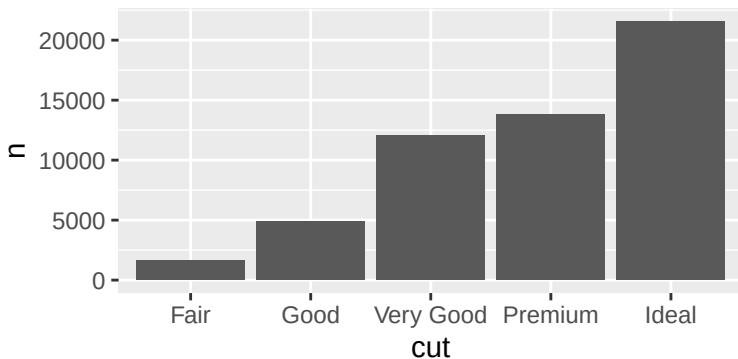
Use `scales::percent()` to add %.

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price)) %>%  
  ungroup() %>%  
  mutate(prop = scales::percent(n/sum(n)))
```

```
## # A tibble: 5 x 4  
##   cut          n price_avg prop  
##   <ord>      <int>    <dbl> <chr>  
## 1 Fair       1610     4359. 3.0%  
## 2 Good       4906     3929. 9.1%  
## 3 Very Good 12082     3982. 22.4%  
## 4 Premium   13791     4584. 25.6%  
## 5 Ideal     21551     3458. 40.0%
```

Graphing after transformation

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price)) %>%  
  ggplot() +  
  geom_bar(aes(x = cut, y = n), stat = "identity")
```



ggplot

Here we used functions from “ggplot2” package. Same pattern as “tidyverse”, but using “+” to connect.

How to write?

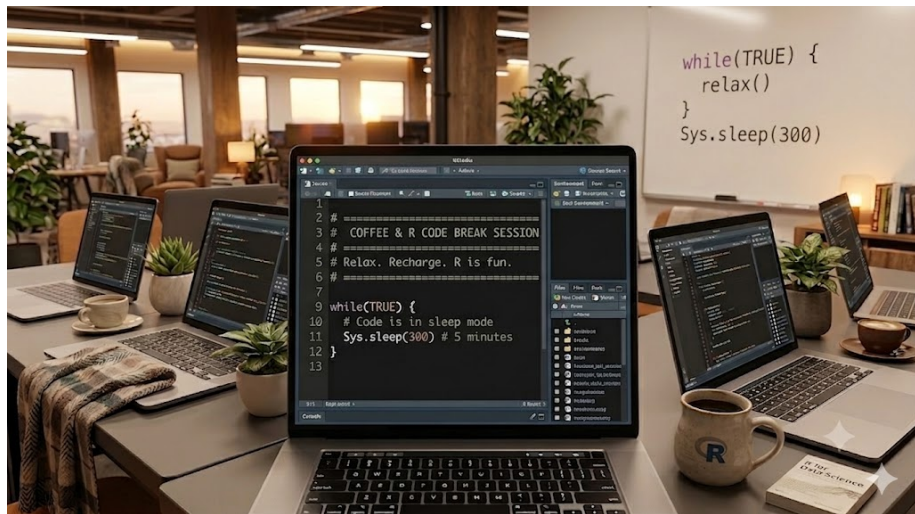
- Specify the data using `ggplot(data = diamonds)`
- Specify the x-/y-axis, `ggplot(data = diamonds, mapping = aes(x = cut))`
- Specify the types of plots with `geom`, e.g. `+ geom_bar()`

```
ggplot(data = diamonds) +  
  geom_bar(mapping = aes(x = cut))
```

More plots

- `geom_histogram()`, `geom_density()`, `geom_line()`, `geom_point()`
- `geom_facet()` generates subplots
- color package
 - RColorBrewer
 - ggsci
- formatting package
 - ggthemes

Break



Install and load package

From what we have learned so far, how to install package? What packages are we going to use?

Install and load package

From what we have learned so far, how to install package? What packages are we going to use?

```
install.packages("palmerpenguins")
```

```
library(tidyverse)  
library(palmerpenguins)
```

Skim the data

How many observations? How many variables? What type?

Skim the data

How many observations? How many variables? What type?

```
glimpse(penguins)
```

```
## Rows: 344
## Columns: 8
## $ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel-
## $ island        <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse-
## $ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
## $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
## $ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186-
## $ body_mass_g   <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
## $ sex           <fct> male, female, female, NA, female, male, female, male-
## $ year          <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

Scatter plot

Consider a scatter plot of flipper length and body mass for `species = "Adelie"`

First let's prepare the data to plot (Hint: `filter`).

Scatter plot

Consider a scatter plot of flipper length and body mass for species = "Adelie"

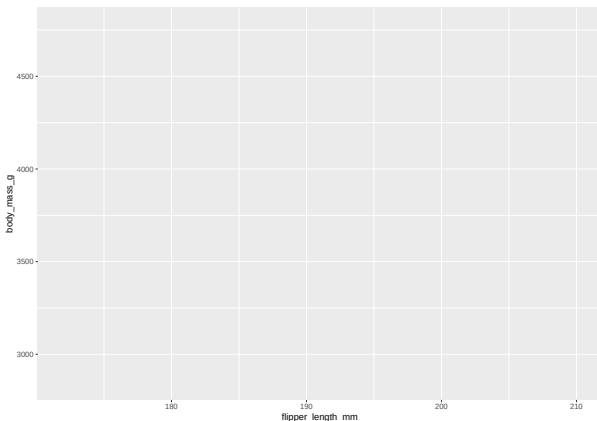
First let's prepare the data to plot (Hint: filter).

```
pdata <- penguins %>%  
  filter(species == "Adelie")
```

A blank canvas

`aes` stands for aesthetic and tells `ggplot` the main characteristics of your plot (x, y, and if the color or fill vary by group)

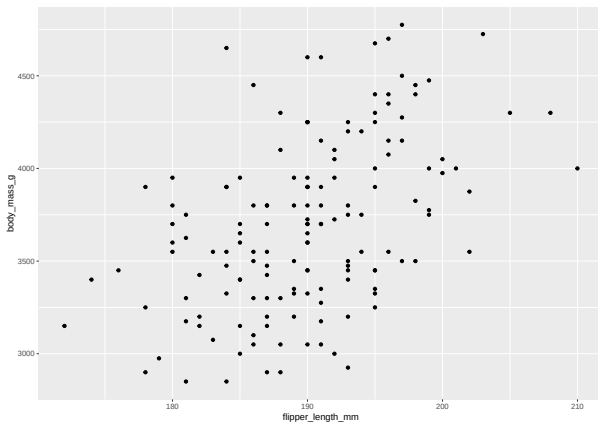
```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g))
```



Add the points

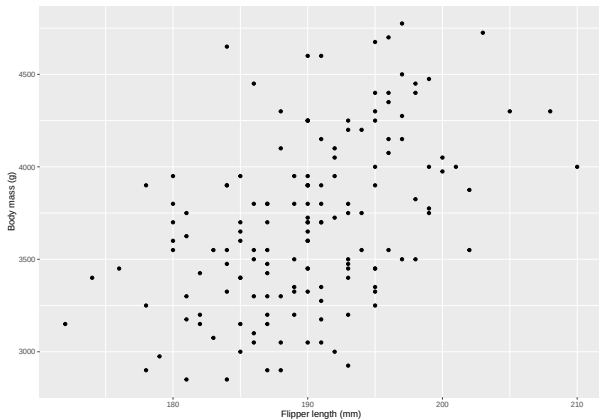
Add layers with ggplot using the +

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point()
```



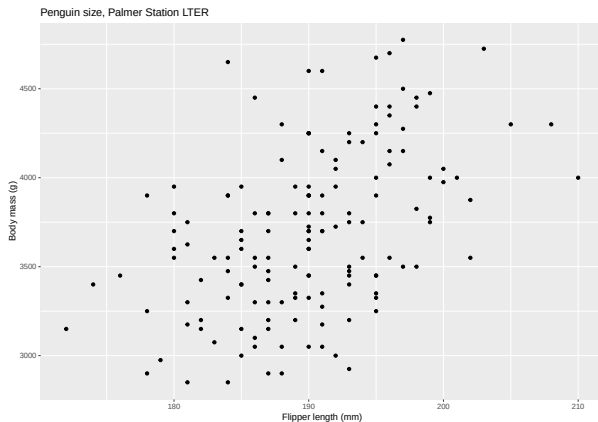
Tidy up labels

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point() +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)")
```



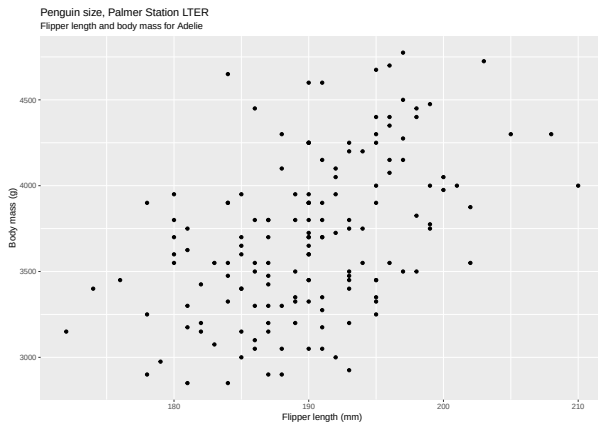
Add a title

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point() +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER")
```



Subtitle

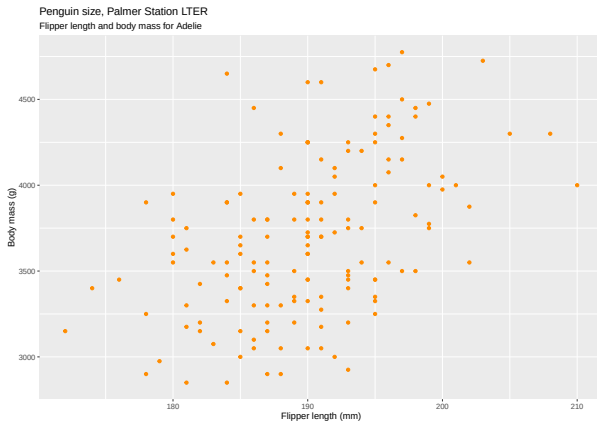
```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point() +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie")
```



Change color of points

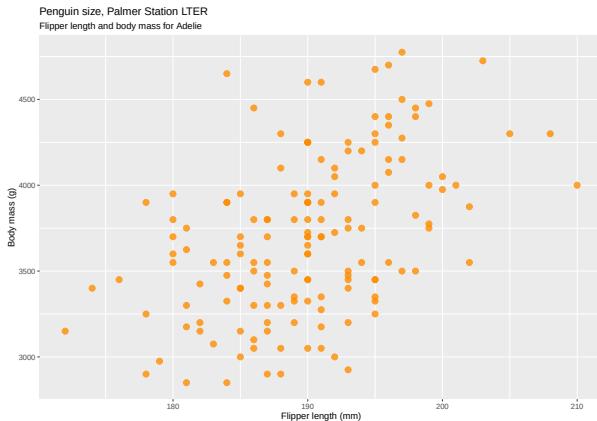
To see all colors, type `colors()`

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(color = "darkorange") +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
        subtitle = "Flipper length and body mass for Adelie")
```



Size of points

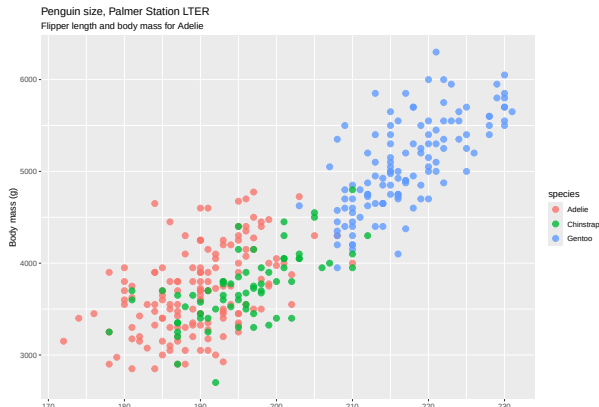
```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(color = "darkorange",  
            size = 3,  
            alpha = 0.8) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie")
```



Coloring by group

Now use the whole data penguins. Specifying the color in `aes()` because it **depends on the data**.

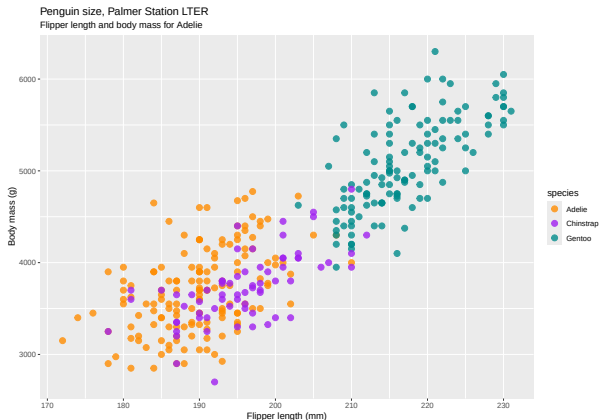
```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
        subtitle = "Flipper length and body mass for Adelie")
```



Change the color manually by group

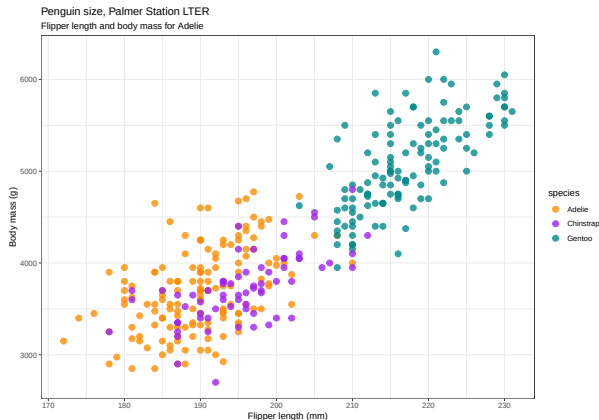
Specifying the color in `aes()` because it **depends on the data**.

```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie")
```



Change the theme

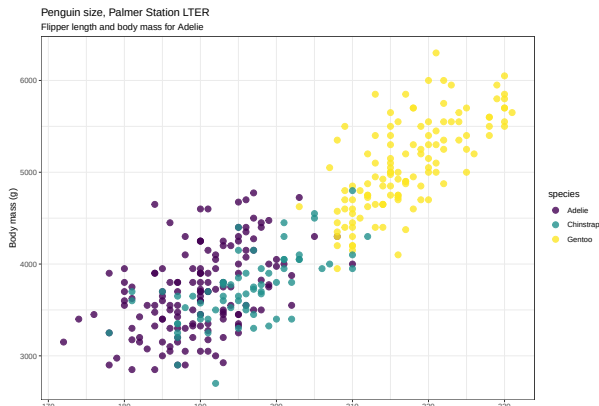
```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie") +  
  theme_bw()
```



Change the color scheme

Optional: try ggsci functions.

```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
        subtitle = "Flipper length and body mass for Adelie") +  
  theme_bw() +  
  scale_color_viridis_d()
```

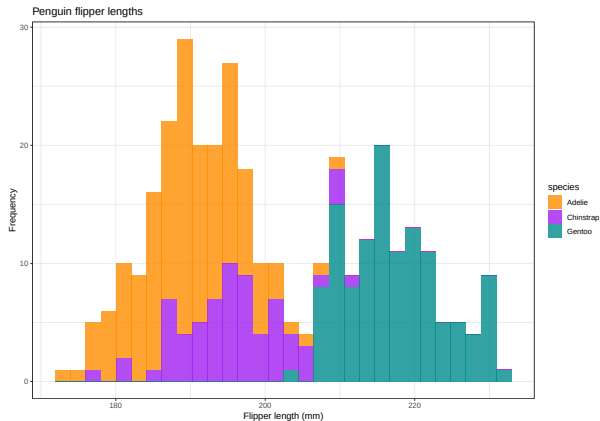


Other plot types?

Other types are also available, e.g. histograms, bar charts, box plots, line graphs and scatter plots.

Histograms

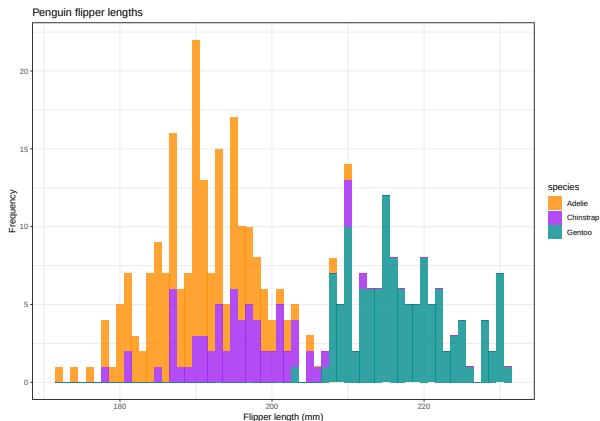
```
ggplot(data = penguins, aes(x = flipper_length_mm, fill = species)) +  
  geom_histogram(alpha = 0.8) +  
  scale_fill_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Frequency") +  
  labs(title = "Penguin flipper lengths") +  
  theme_bw()
```



Histogram

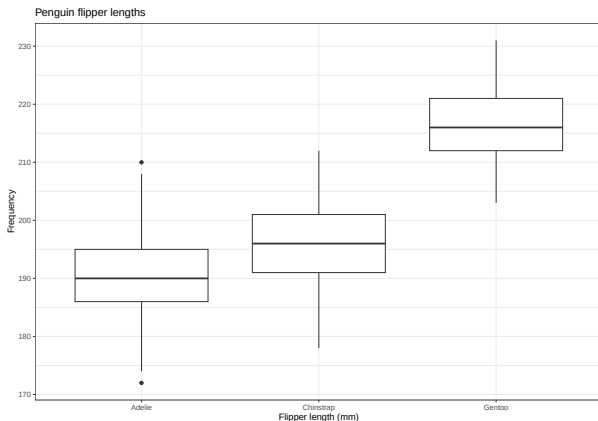
Try `binwidth =` and `bins ==`

```
ggplot(data = penguins, aes(x = flipper_length_mm, fill = species)) +  
  geom_histogram(binwidth = 1, alpha = 0.8) +  
  scale_fill_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Frequency") +  
  labs(title = "Penguin flipper lengths") +  
  theme_bw()
```



Boxplots

```
ggplot(data = penguins, aes(x = species, y = flipper_length_mm)) +  
  geom_boxplot(show.legend = FALSE) +  
  xlab("Flipper length (mm)") +  
  ylab("Frequency") +  
  labs(title = "Penguin flipper lengths") +  
  theme_bw()
```



Barcharts

First, try to summarize the penguin data by species and returns the proportion of each penguin types.

Barcharts

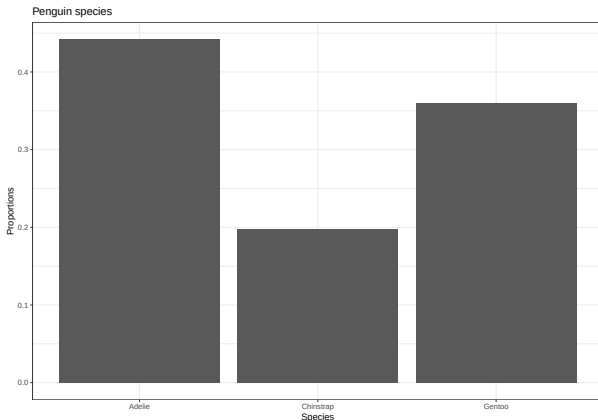
First, try to summarize the penguin data by species and returns the proportion of each penguin types.

```
gplot <- penguins %>%  
  group_by(species) %>%  
  tally() %>%  
  mutate(prop = n / sum(n))  
gplot
```

```
## # A tibble: 3 x 3  
##   species      n prop  
##   <fct>    <int> <dbl>  
## 1 Adelie    152 0.442  
## 2 Chinstrap  68 0.198  
## 3 Gentoo   124 0.360
```

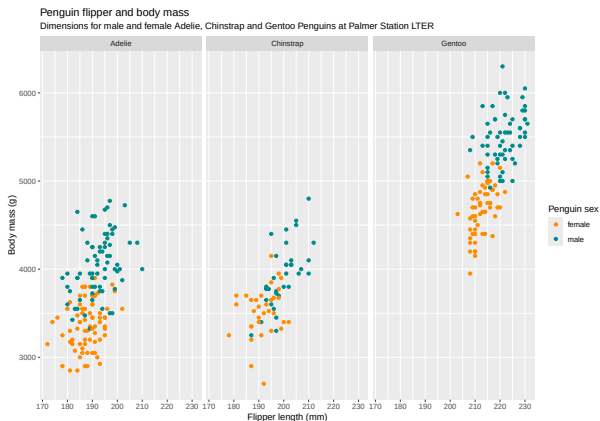
Bar charts

```
ggplot(data = gplot, aes(x = species, y = prop)) +  
  geom_bar(stat = "identity") +  
  xlab("Species") +  
  ylab("Proportions") +  
  labs(title = "Penguin species") +  
  theme_bw()
```

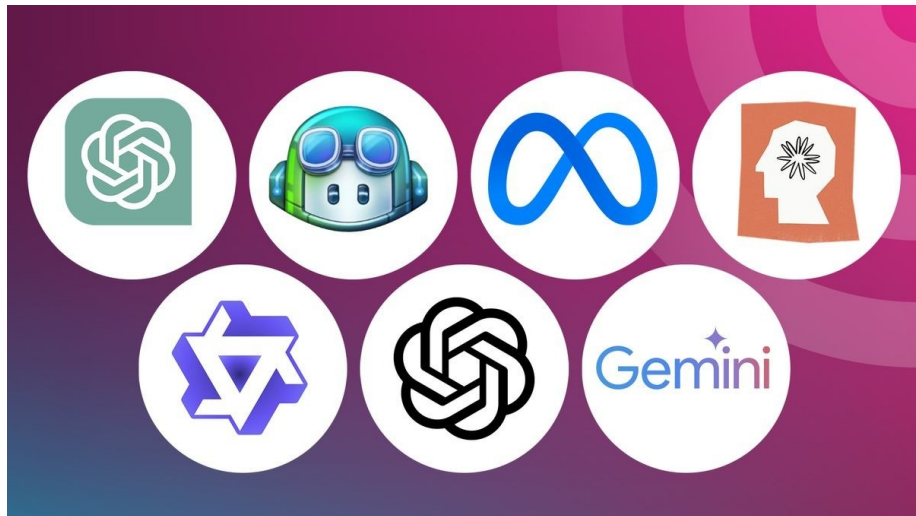


Faceting

```
ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(aes(color = sex)) +  
  scale_color_manual(values = c("darkorange", "cyan4"), na.translate = FALSE) +  
  labs(title = "Penguin flipper and body mass",  
       subtitle = "Dimensions for male and female Adelie, Chinstrap and Gentoo Penguins at Palmer Station LTER",  
       x = "Flipper length (mm)",  
       y = "Body mass (g)",  
       color = "Penguin sex") +  
  facet_wrap(~species)
```



Using LLMs for writing R code



Popular LLMs

Online LLMs

- ChatGPT *
- Claude
- Gemini *
- Grok
- DeepSeek

(*: My preferred choices)

Local LLMs

- Ollama
 - Open source LLMs.
 - Performance depends on the model choice and your machine
 - Secure, but generally not as good as Online LLMs

Important Tips

- Never provide sensitive data to an LLM
- Do not accept LLM responses for absolute truth (they still make mistakes and overcomplicate!)
- YOU are fully responsible for the accuracy, originality, licensing, and ethical/legal compliance of work that you do.
 - Failure to provide a complete and accurate disclosure of AI usage may be considered an ethical breach in research/work contexts.

Example - Pivot Data

Prompt

I have a wide dataset (called table4a) with the "country" column as the id variable and columns named "1999" and "2000" that provide the number of TB cases in each country for that year.

Provide R code that will convert this wide dataset to long form

```
library(tidyr)

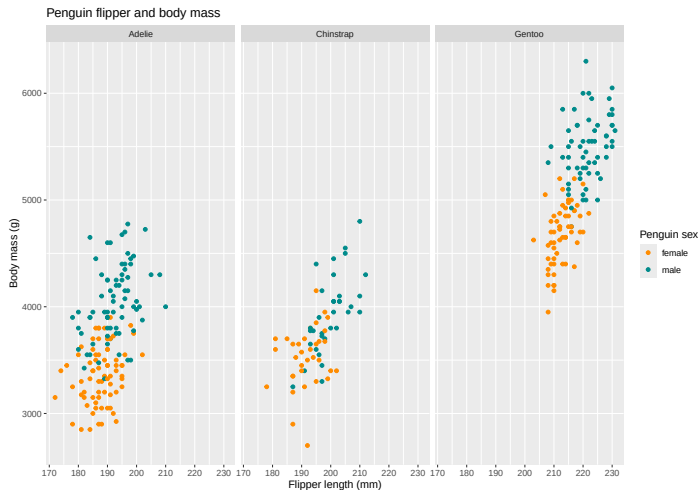
table4a_long <- table4a|>
  pivot_longer(
    cols = c(`1999`, `2000`),
    names_to = "year",
    values_to = "cases"
  )

table4a_long
```

```
## # A tibble: 6 x 3
##   country    year  cases
##   <chr>      <chr> <dbl>
## 1 Afghanistan 1999     745
## 2 Afghanistan 2000    2666
## 3 Brazil      1999   37737
## 4 Brazil      2000  80488
## 5 China       1999 212258
## 6 China       2000 213766
```

Example - Improve Visualizations

Before



Example - Improve Visualizations

After Prompt:

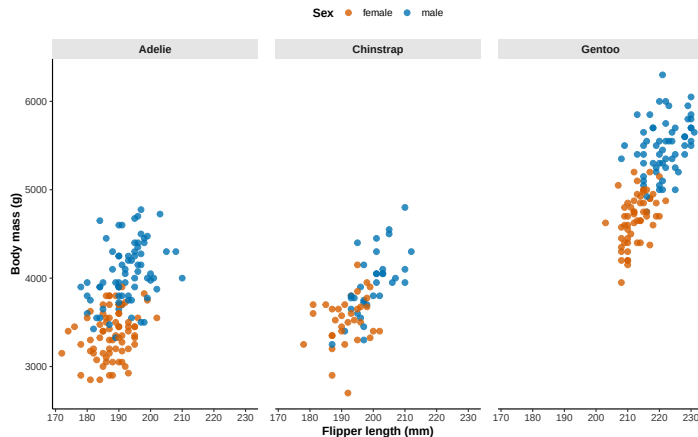
I have the following R code:

```
<PASTE R CODE HERE>
```

Please update the code so that the rendered visual is publication ready

Penguin flipper length and body mass

Measurements for male and female Adelie, Chinstrap, and Gentoo penguins at Palmer Station LTER



Exercises

Available on course website.